

U.S. SEMICONDUCTOR: TRADE & LABOR ANALYSIS

U.S. Semiconductor Exports — Source Data & Cleaning Process | HS Code 3542 | 2025

Overview

This document describes the data cleaning steps applied to the UN Comtrade U.S. semiconductor exports dataset (HS 3542 Un Comtrade USA Exports.csv). The objective was to simplify the dataset so that it focuses on variables that actually matter for analyzing U.S. semiconductor exports — removing redundant or misleading fields, converting critical columns to numeric types, and producing a file that is lean, transparent, and ready for budgetary or trade-partner analysis. The cleaned output is saved as Cleaned_HS_3542_Un_Comtrade_USA_Exports.csv.

Step 1 — Reading the Raw Dataset

Objective

Load the original UN Comtrade export dataset into a pandas DataFrame for inspection and transformation.

Code

```
df = pd.read_csv('HS 3542 Un Comtrade USA Exports.csv', encoding='Latin1')
```

Key Design Choice

- **Latin1 encoding:** Specified to ensure proper handling of special characters in country names (e.g., Turkiye, Curacao). The raw file contained many fields, some redundant or consistently empty.

Step 2 — Converting Key Columns to Numeric

Objective

Ensure that partner codes, commodity codes, quantity fields, and trade values are correctly typed before any calculations or aggregations.

Code

```
for col in ['partnerCode', 'cmdCode', 'qty', 'qtyUnitCode',  
           'altQtyUnitCode', 'altQty', 'fobvalue', 'primaryValue']:  
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

Key Design Choice

- **errors='coerce':** Safely converts invalid entries to NaN rather than raising an exception, allowing the analysis to continue while flagging problematic values.

Step 3 — Dropping Redundant or Misleading Columns

Objective

Remove columns that add no analytical value, are consistently empty, duplicate existing information, or introduce unnecessary complexity.

Code

```
cols_to_drop = [  
    'flowCode', 'partner2Code', 'partner2ISO', 'mosCode', 'motCode',
```

```

'netWgt', 'period', 'grossWgt', 'legacyEstimationFlag', 'cifvalue',
'primaryValue', 'altQtyUnitCode', 'altQtyUnitAbbr', 'altQty',
'isAltQtyEstimated', 'motDesc', 'customsDesc', 'isGrossWgtEstimated',
'isNetWgtEstimated', 'qty', 'qtyUnitCode', 'isAggregate',
'cmdCode', 'partnerCode', 'reporterISO', 'isReported',
'partner2Desc', 'isQtyEstimated'
]
df = df.drop(columns=cols_to_drop)

```

Rationale for Each Dropped Column Group

Column(s)	Reason for Removal
partner2Code, partner2ISO, partner2Desc, mosCode	No analytical value for bilateral U.S. export analysis
netWgt, grossWgt	Consistently zero or empty in this dataset — misleading if retained
primaryValue	Duplicates fobvalue; no additional information
altQty, altQtyUnitCode, altQtyUnitAbbr, isAltQtyEstimated	Duplicates qty; consistently redundant
motCode, motDesc, customsDesc	Not detailed enough to support meaningful logistics analysis
cifvalue, legacyEstimationFlag	Empty in export records (CIF applies to imports)
reporterISO, isReported, isAggregate, isGrossWgtEstimated, isNetWgtEstimated	Metadata fields not needed for trade value analysis
period, flowCode, cmdCode, partnerCode, qtyUnitCode, isQtyEstimated, qty	Redundant with other retained fields or not meaningful for this analysis

Step 4 — Replacing Placeholder Values

Objective

Standardize the reporter country name to ensure consistency and avoid mislabeling in country-level analysis.

Code

```
df['reporterDesc'] = df['reporterDesc'].replace('USA', 'United States')
```

Key Design Choice

- **'USA' to 'United States':** The raw dataset used the abbreviation 'USA' as the reporter country identifier. Standardizing to 'United States' ensures consistency with partner country naming conventions and avoids grouping errors in downstream analysis.

Step 5 — Saving the Cleaned Dataset

Objective

Export the cleaned dataset as a structured CSV file ready for EDA, SQL loading, and visualization.

Code

```
df.to_csv('Cleaned_HS_3542_Un_Comtrade_USA_Exports.csv', index=False)
```

Output Summary

Metric	Value
Output file	Cleaned_HS_3542_Un_Comtrade_USA_Exports.csv
Rows retained	950
Key fields retained	reporter, partner, partnerISO, refMonth, refYear, fobvalue, cmdDesc, aggrLevel
Reporter label	Standardized to 'United States'

Step 6 — Correcting Partner Names in Excel

Objective

Normalize country names in the partnerDesc column to correct common inconsistencies introduced by UN Comtrade's labeling conventions.

Excel Formula

```
=SUBSTITUTE (
  SUBSTITUTE (
    SUBSTITUTE (
      SUBSTITUTE (
        SUBSTITUTE (
          SUBSTITUTE (
            SUBSTITUTE (M2,
              "Bolivia (Plurinational State of)", "Bolivia"),
              "Dominican Rep.", "Dominican Republic"),
              "China, Hong Kong SAR", "China, Hong Kong"),
              "Rep. of Korea", "South Korea"),
              "China, Macao SAR", "China, Macao"),
              "Saint Vincent and the Grenadin", "St. Vincent and the Grenadines"),
              "Saint Kitts and Nevis", "St. Kitts and Nevis")
```

Corrections Applied

Original Name	Standardized Name
Rep. of Korea	South Korea
Dominican Rep.	Dominican Republic
Bolivia (Plurinational State of)	Bolivia
China, Hong Kong SAR	China, Hong Kong
China, Macao SAR	China, Macao
Saint Vincent and the Grenadin	St. Vincent and the Grenadines
Saint Kitts and Nevis	St. Kitts and Nevis

Key Decisions and Rationale

- **Weights dropped:** netWgt and grossWgt were always zero in this dataset, so retaining them would have introduced misleading interpretations.
- **Logistics ignored:** Transport mode (motCode, motDesc) was not detailed enough to support meaningful logistics analysis — focus was placed on value and partner data.
- **Redundancy reduced:** Columns like primaryValue and altQty duplicated existing information and were eliminated for simplicity and clarity.
- **Placeholder fixed:** Replacing 'USA' with 'United States' ensures clarity and avoids mislabeling in country-level analysis.
- **Focus sharpened:** Aggressive cleaning ensures the dataset is ready for budgetary analysis and partner comparisons — the most relevant insights for this project.

Data Source

[UN Comtrade](#) — HS 3542 Un Comtrade USA Exports.csv